

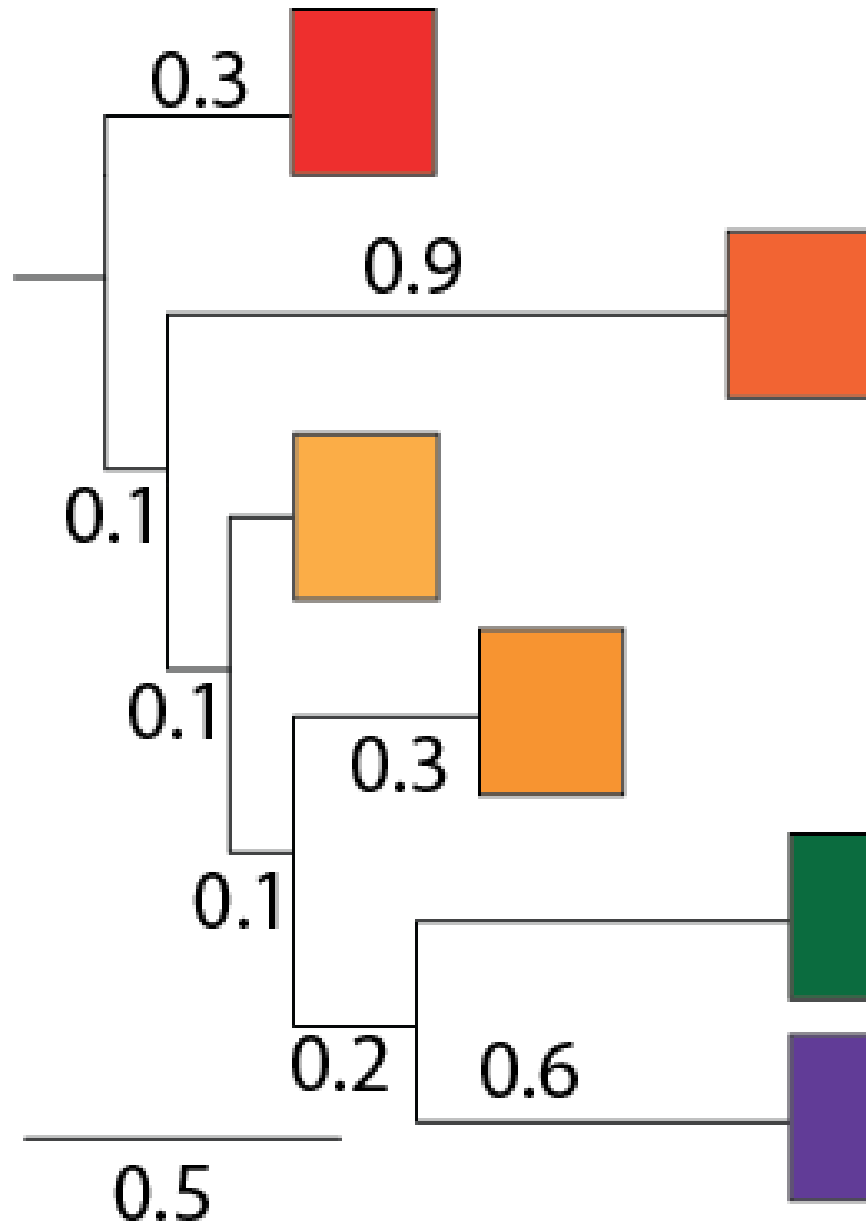
Phylogenetic inference of epidemiological processes

Evolutionary and epidemiological mechanisms determine the patterns of spread, emergence and persistence of pathogens

Questions this lecture addresses

- What processes generate and redistribute pathogen diversity?
- Which features of those processes can appear in a phylogeny?
- Which mechanisms can produce similar tree patterns?
- What additional data make a molecular epidemiologic inference credible?

Dimensions



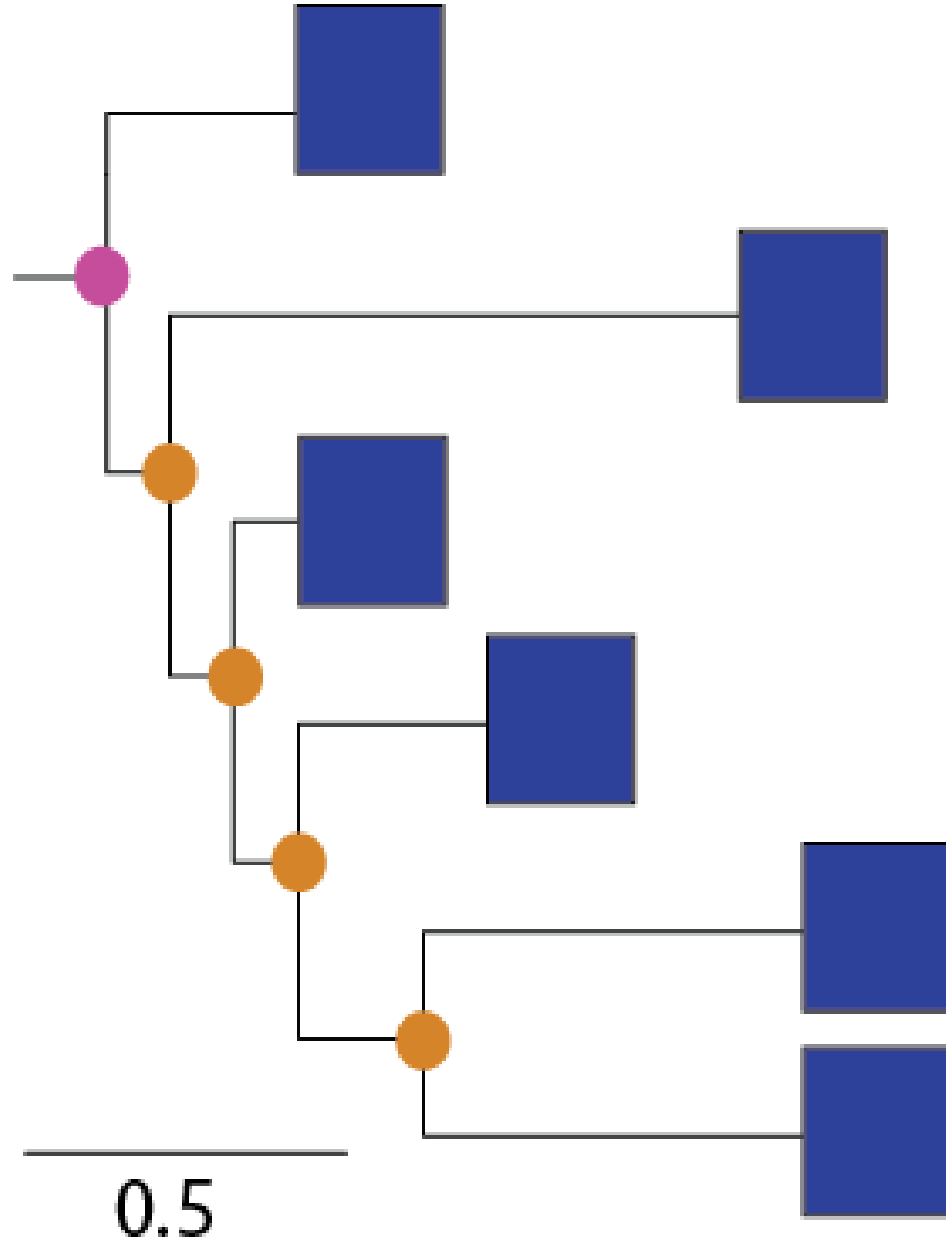
Vertical dimension has no meaning

Horizontal dimension gives the amount of genetic change

Horizontal lines show evolutionary changes in lineages over time (i.e. branch lengths are informative)

Scale bar shows the length of branch equivalent to 0.5 nucleotide substitutions/site

Structure



Tip/Taxon/Virus/Leave
-Sequenced Viruses
-we also know date and
location of isolation,
host and clinical features

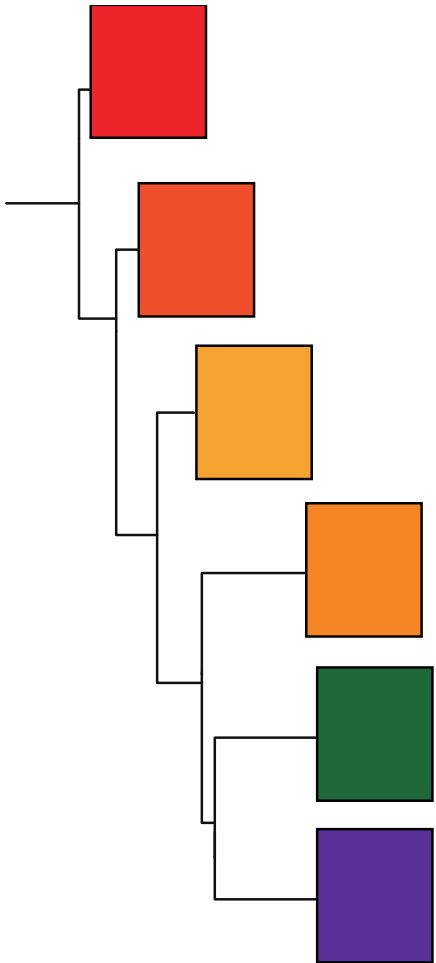


Internal Node
Putative common
Ancestor



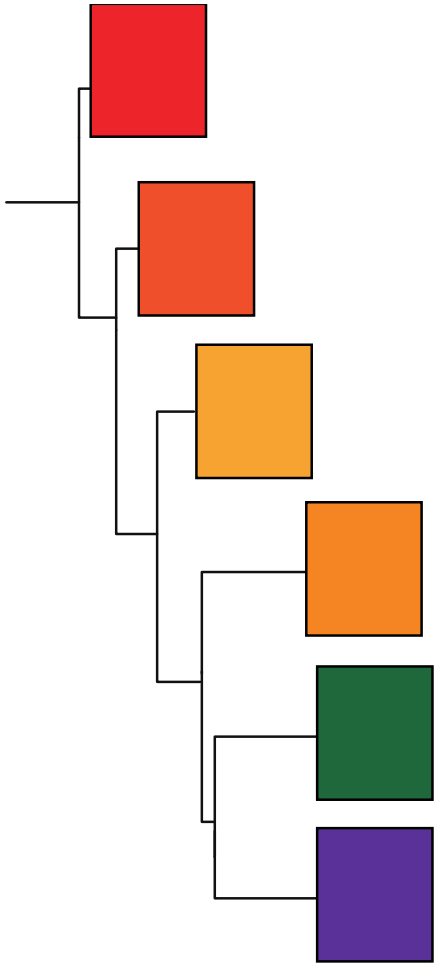
Root – provides an
order for sequence of
events leading to
observation (sequenced
virus)

Notes on phylogenies



- A phylogeny is an estimate, not an observed object.
 - Weakly supported nodes, polytomies, rooting choices, and model assumptions can change the interpretation.
- Branch length may represent genetic change or calendar time, depending on the tree.
- Genetically similar samples may share an epidemiological connection, but proximity alone does not establish direct transmission, direction, or an immediate source.

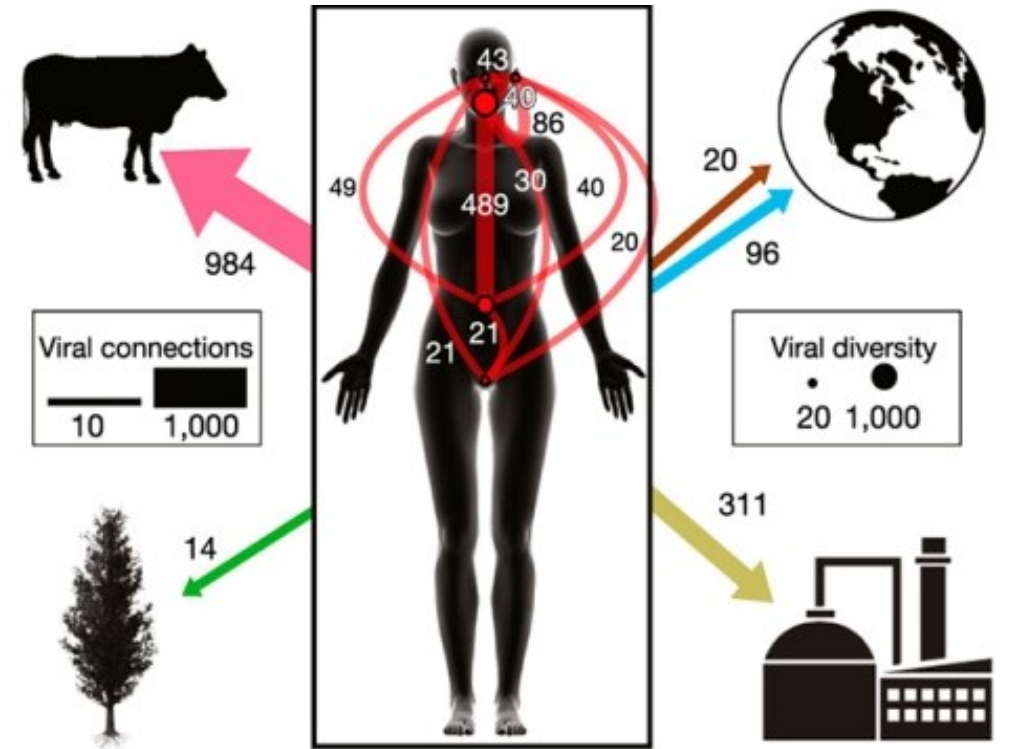
Notes on phylogenies



- Basal placement depends on how the tree is rooted. An early or basal sample is not necessarily the source.
- Cluster definitions depend on the sampling frame, genomic region, evolutionary rate, analysis method, and threshold. Inclusion does not prove linkage, and exclusion does not eliminate it.

The Modern Era – Systems and System Interactions

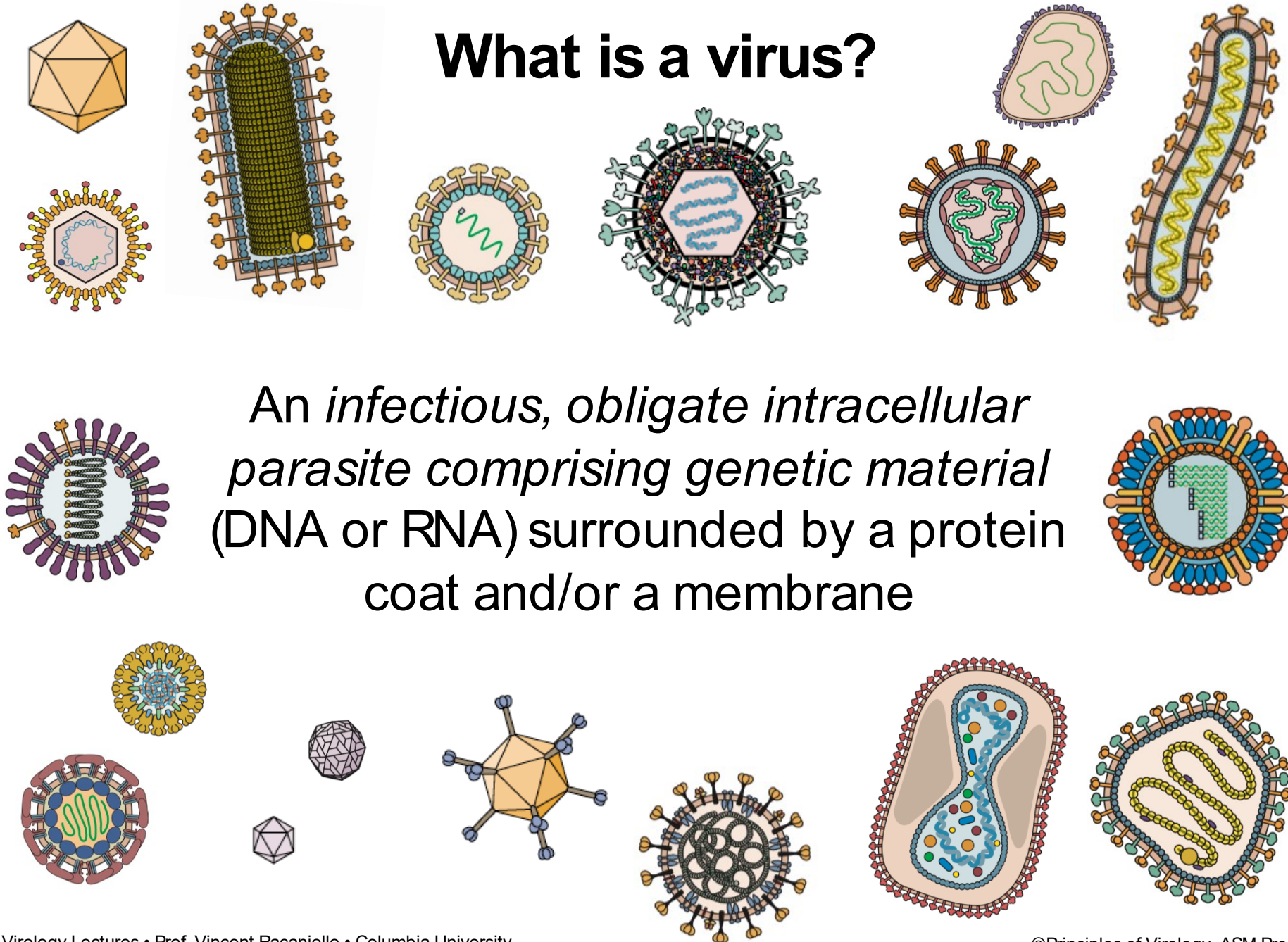
Habitat distribution of metagenomic viruses



D Paez-Espino *et al.* *Nature* 1–6 (2016) doi:10.1038/nature19094

nature

What is a virus?

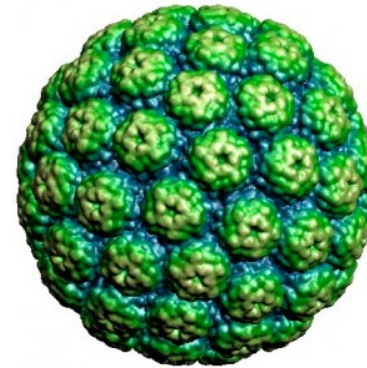


An infectious, obligate intracellular parasite comprising genetic material (DNA or RNA) surrounded by a protein coat and/or a membrane

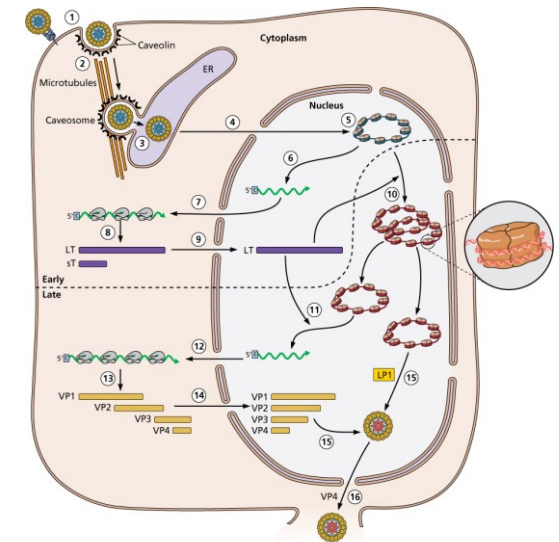
- The virion contains genetic information that is fundamental to complete the life cycle of the virus
- The genetics of the virion can reveal insights into the mechanisms needed to complete the life cycle
- Studying populations of virions can reveal insights into the population of infected hosts.

The virus and the virion

A virus is an organism with two phases



virion



infected cell

Despite enormous viral diversity

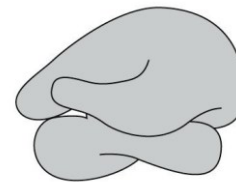
- **FINITE NUMBER OF VIRAL GENOME TYPES.**



- A simplified virus classification system based on mechanism of viral replication

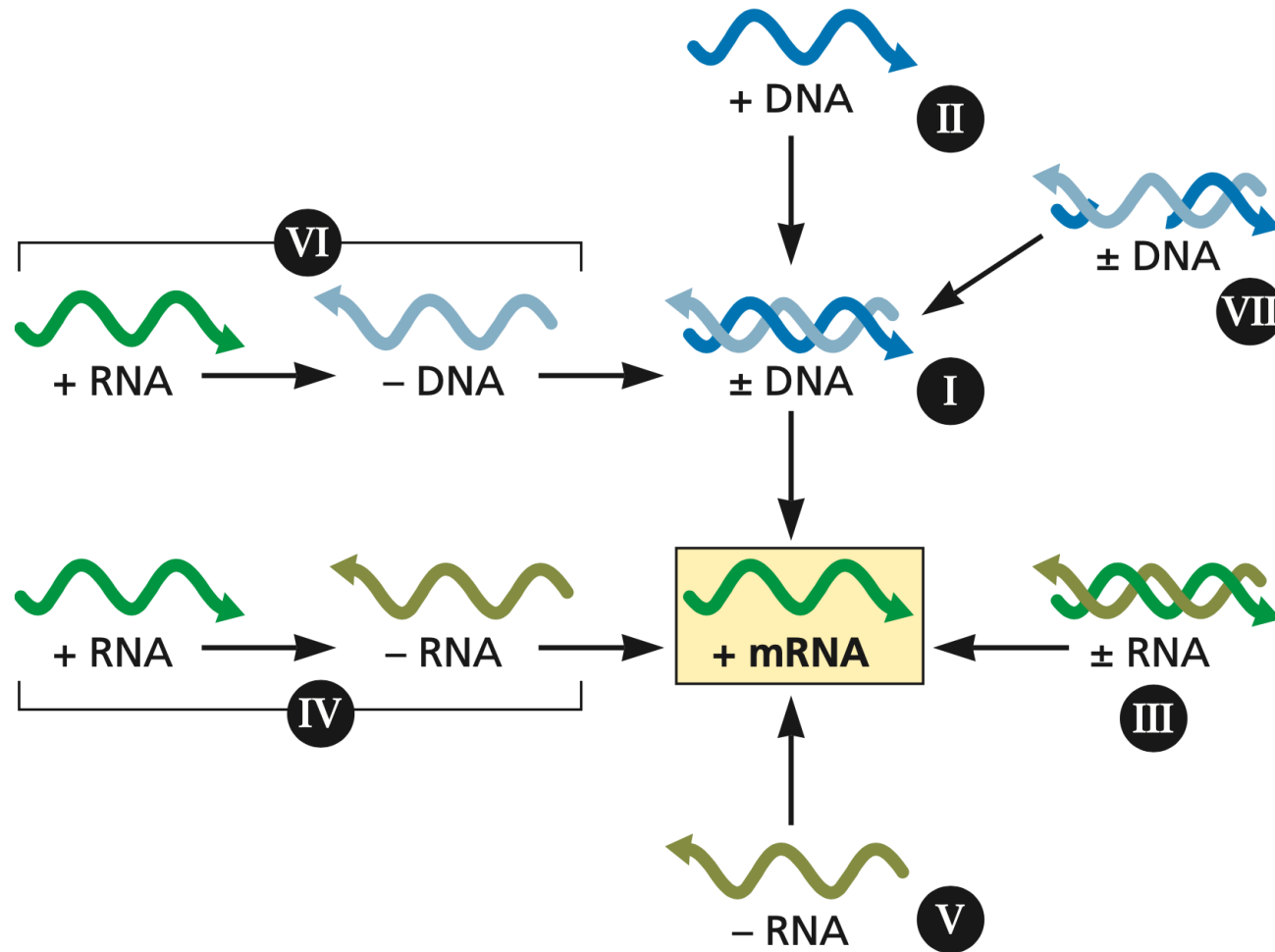
Key fact makes your life easier:

Viral genomes must make mRNA that can be read by host ribosomes



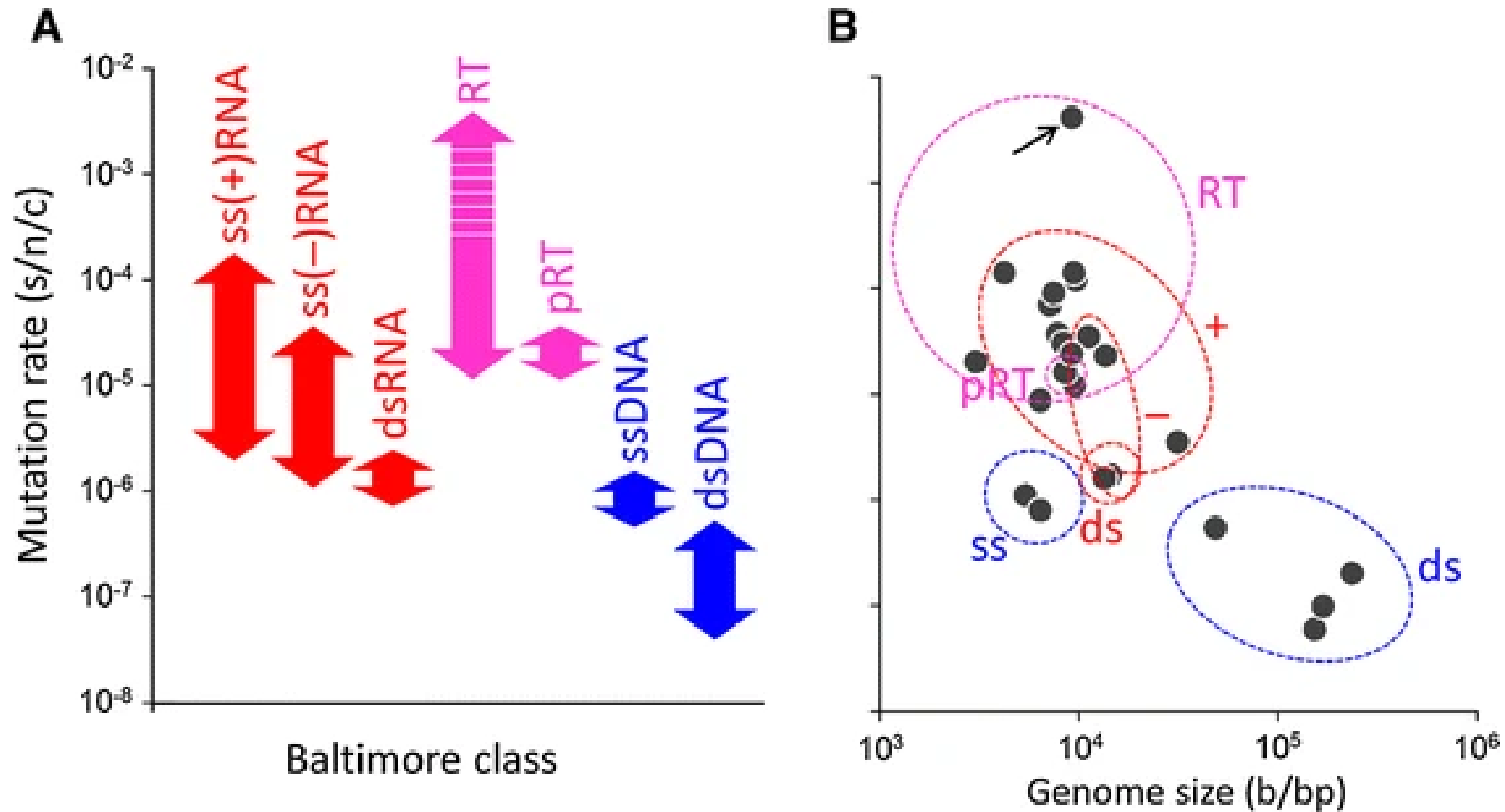
All viruses on the planet follow this rule, no known exception

Described by David Baltimore (Nobel Laureate)



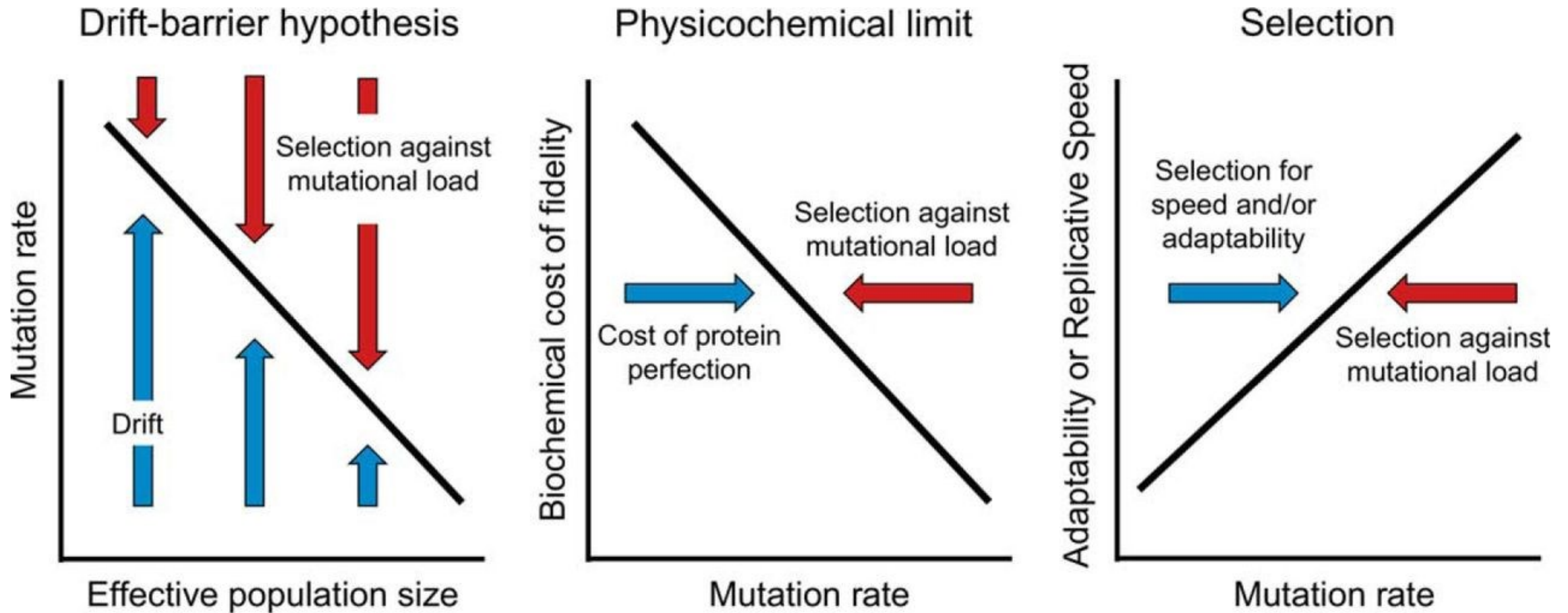
The original Baltimore system missed one genome type:
gapped DNA of *Hepadnaviridae*

Mechanisms of viral mutation



Most mutations are deleterious

- Mutations create genetic variants whose effects depend on the environment and genetic background.
- Most new mutations are neutral or deleterious; relatively few increase fitness.
- Why have pathogens not evolved toward a zero mutation rate?



Mutation rate is not an evolutionary rate

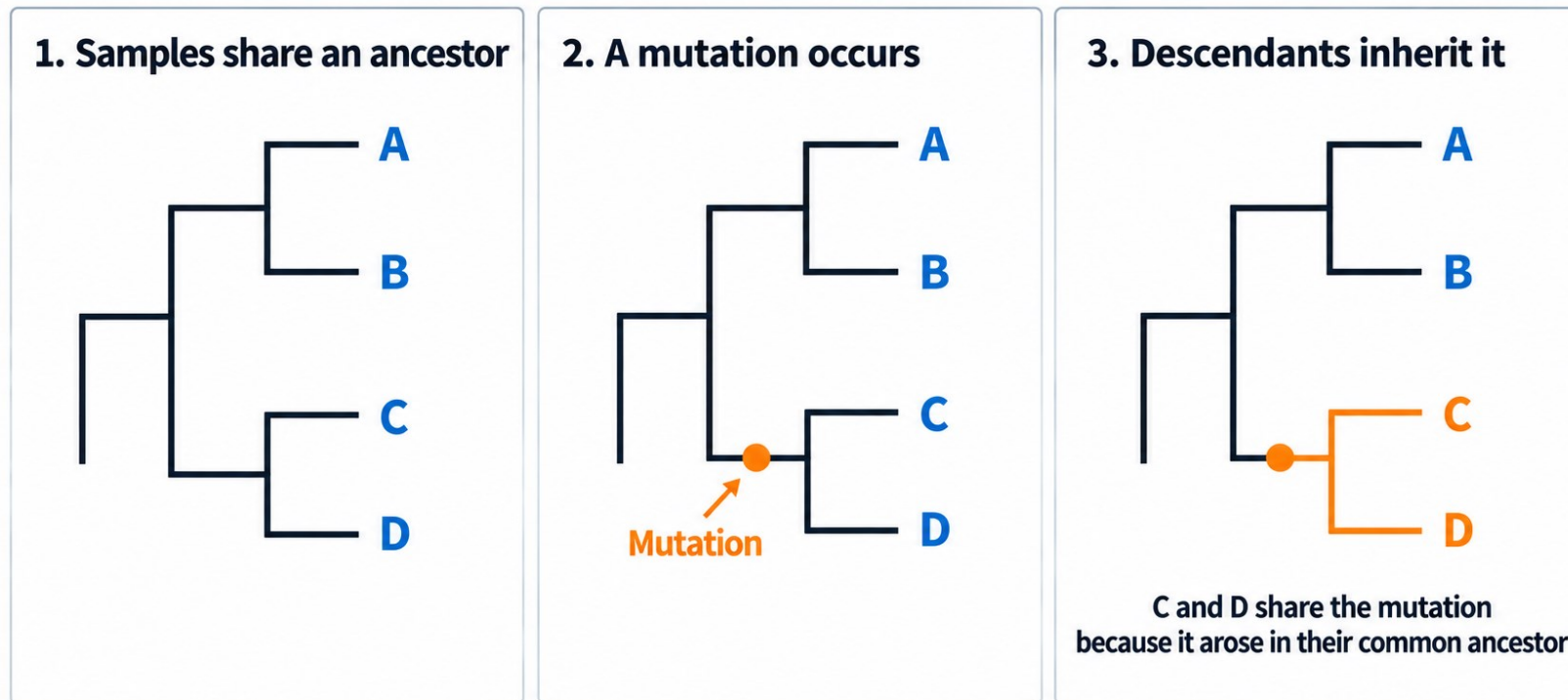
- Mutation rate measures new errors per replication event or unit time.
- Substitution rate measures changes that survive drift, selection, transmission, and sampling.
- The molecular clock is therefore a population-level outcome, not a direct polymerase readout.
- Compare rates only after specifying the denominator, genomic region, and timescale.

Evolutionary processes act on diversity in different ways

- Mutation and genome exchange generate variants.
 - Selection and drift change their frequencies.
 - Migration redistributes variants among populations.
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- When we sequence a population of viruses we are observing genetic diversity resulting from those processes.

Diversity is the output of a causal process

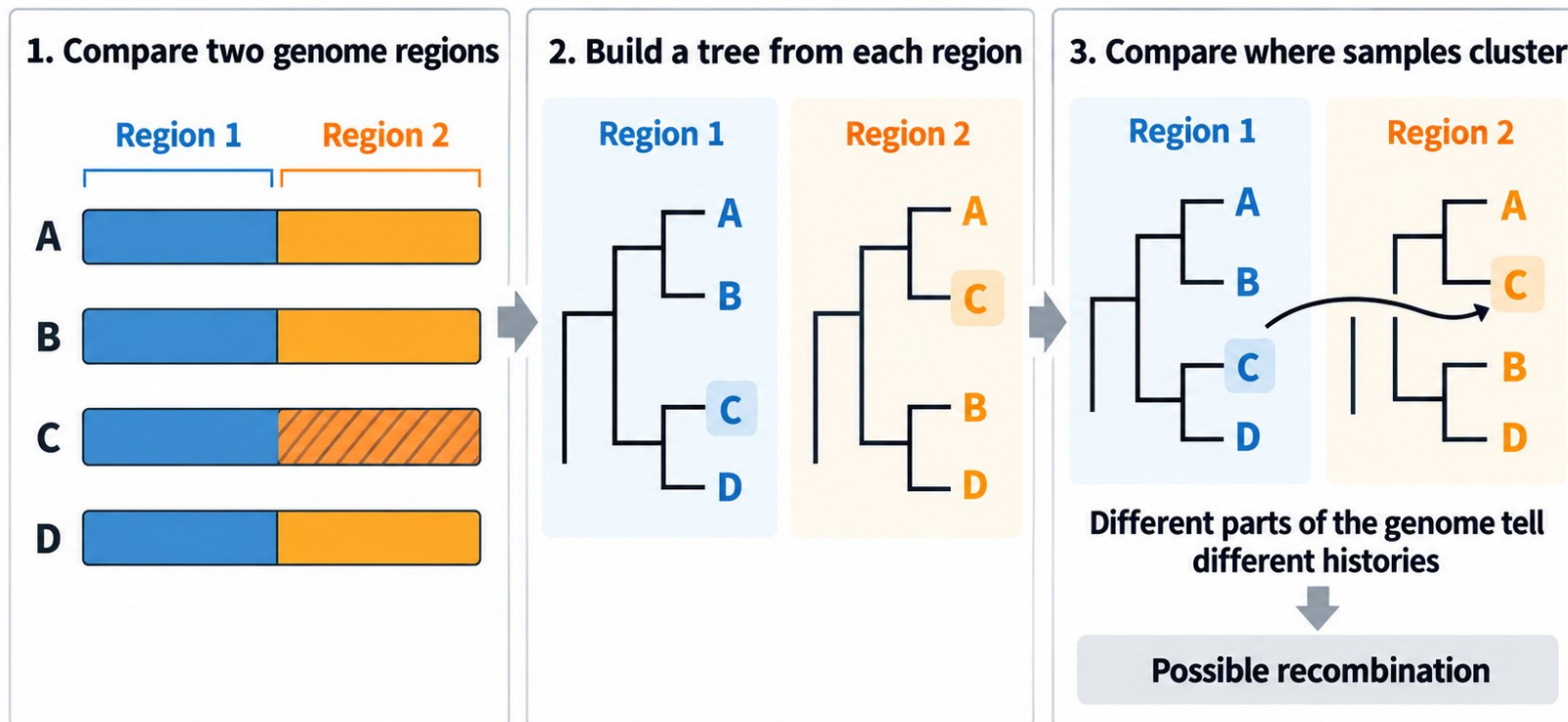
- Generation: mutation, recombination, and reassortment create variants



A mutation marks a change along a branch—not a change at every split.

Diversity is the output of a causal process

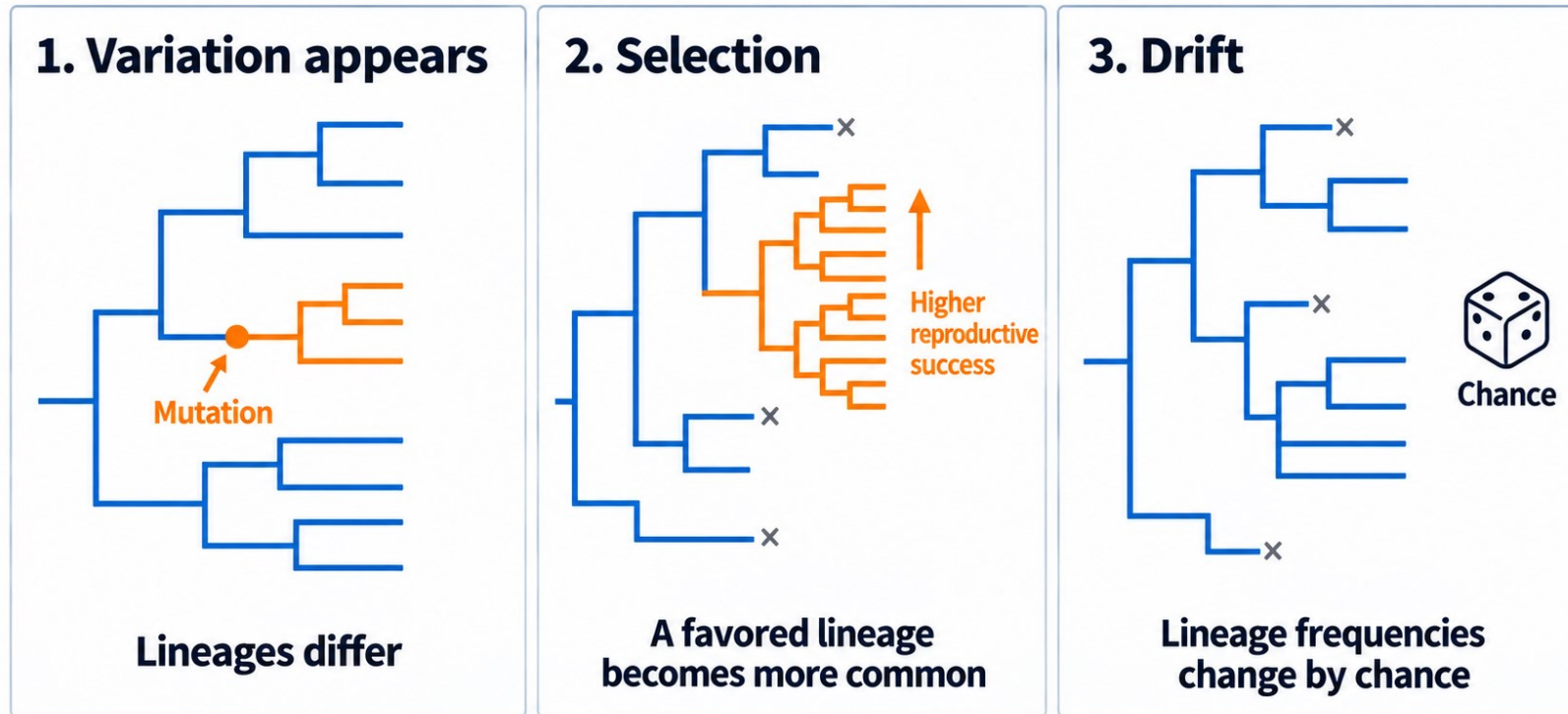
- Genome exchange reshapes diversity
 - Phylogenetic incongruence can be signal, not analytical failure.



Tree disagreement is a clue, not proof: also consider low signal, sequencing error, and model choice.

Diversity is the output of a causal process

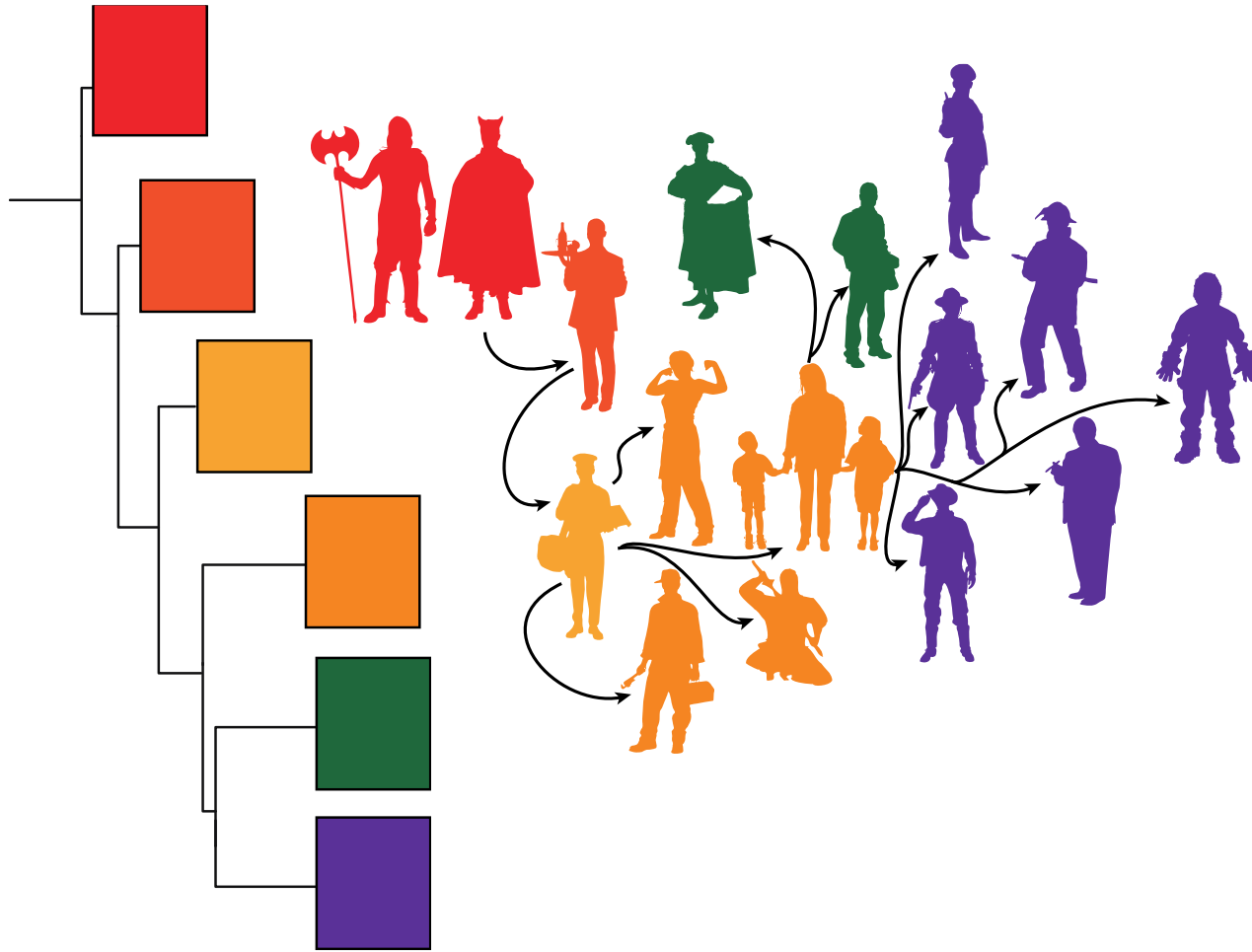
- Filtering: selection and drift change frequencies within hosts and between hosts.



Selection is directional; drift is stochastic. Tree shape alone does not prove the cause.

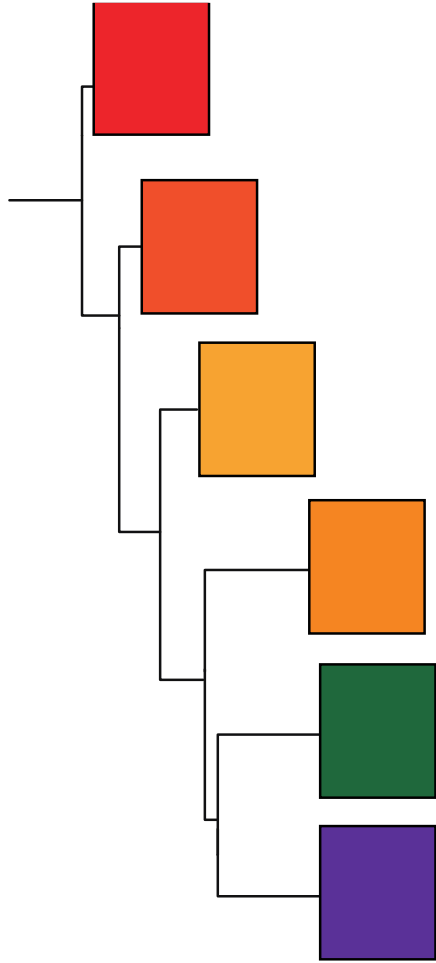
The observed phylogeny is a filtered record

- Observation: surveillance and sequencing sample the resulting population imperfectly.
- Evolutionary processes generate variation.
- Transmission moves only part of that variation.
- Surveillance samples only part of what circulated.



Molecular epidemiology combines genomes with time, location, host, exposure, and surveillance data to distinguish among these explanations.

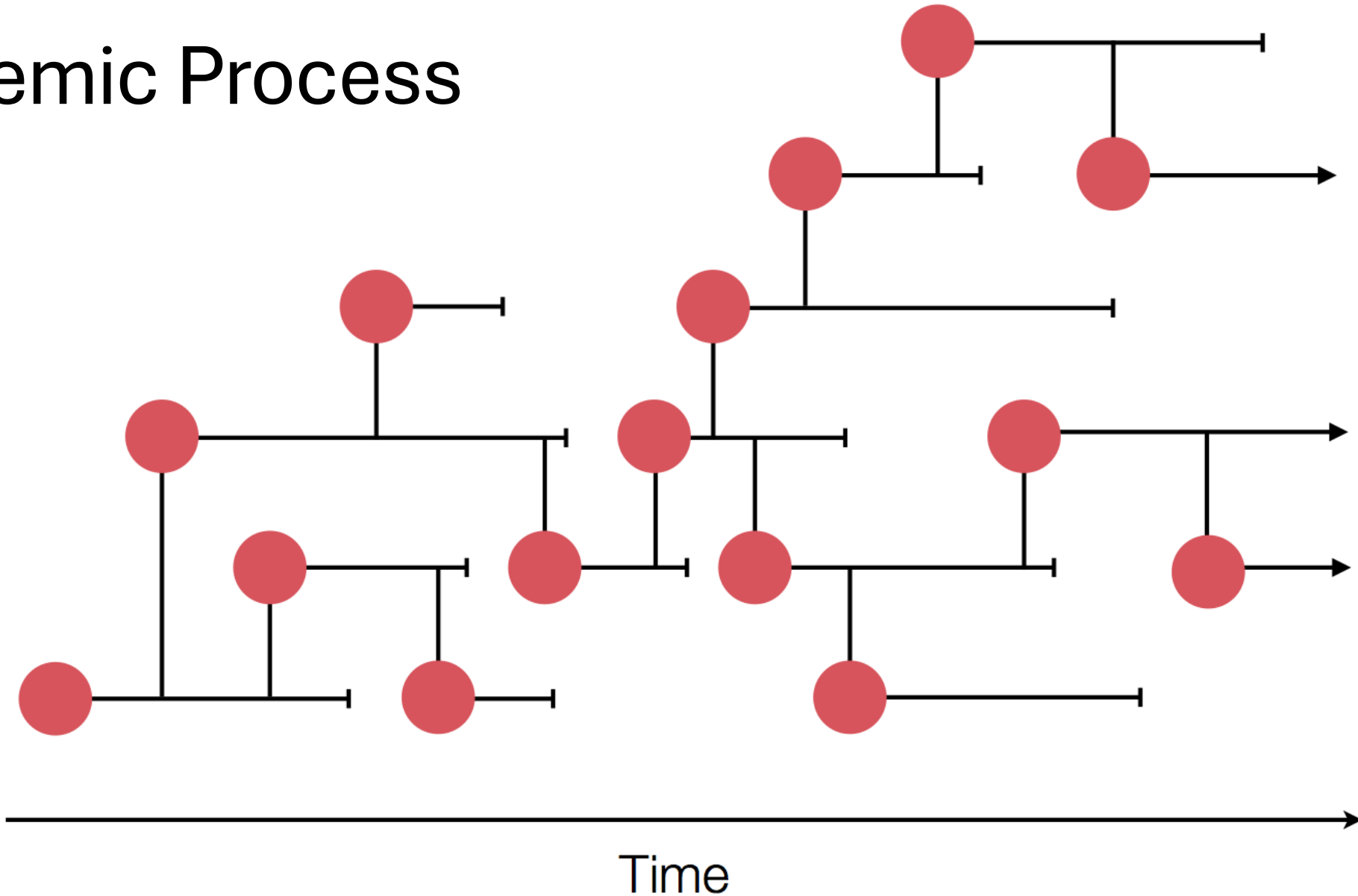
Phylodynamics



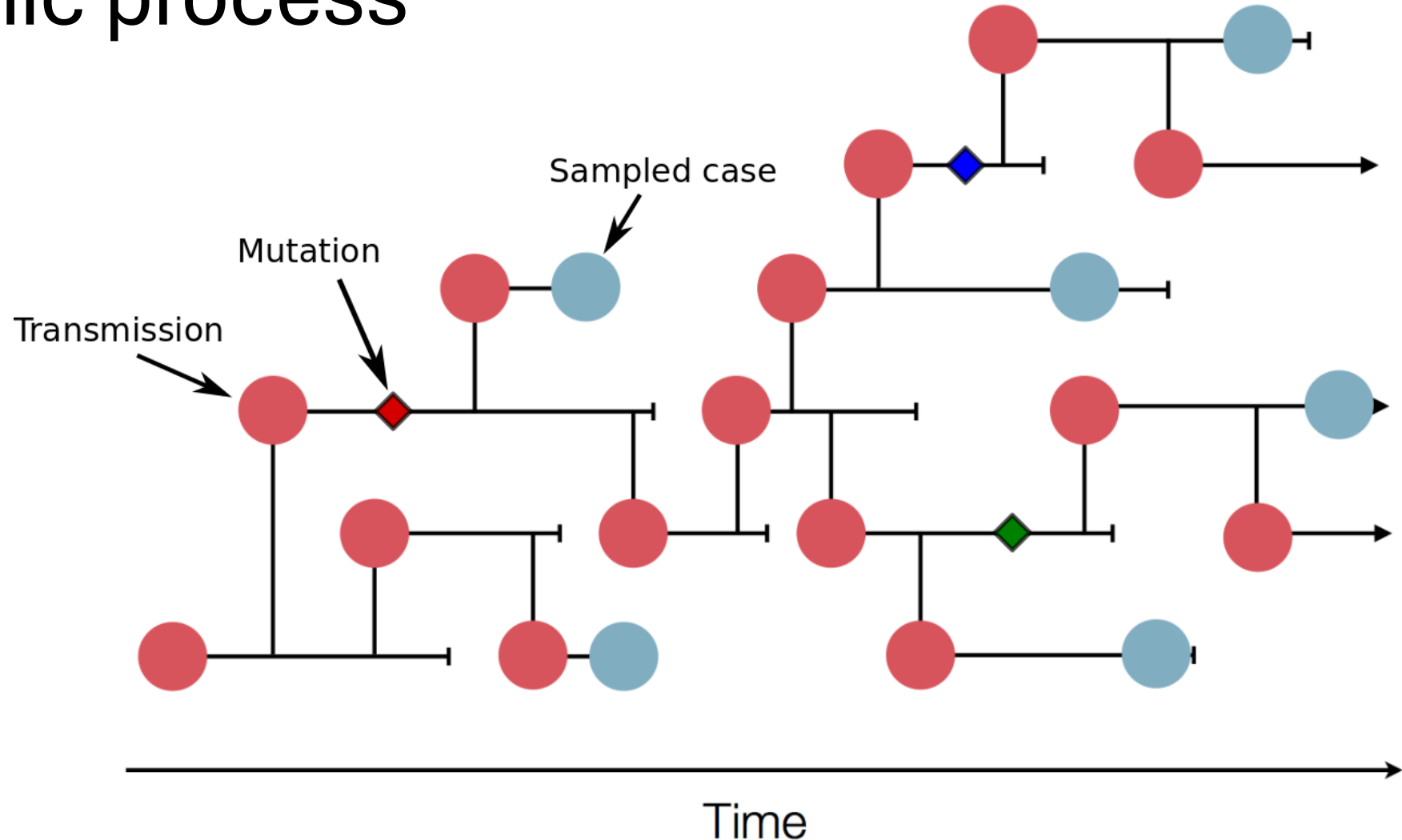
The rapid evolution of many pathogens, particularly RNA viruses, means that their evolution and ecology occur on the same timescale, and therefore must be studied jointly to be fully understood.

- Samples included in a phylogenetic analysis are modeled as related through common ancestry.
- Small genetic distances indicate recent shared ancestry under the assumed model.
- They do not establish direct transmission.

Epidemic Process

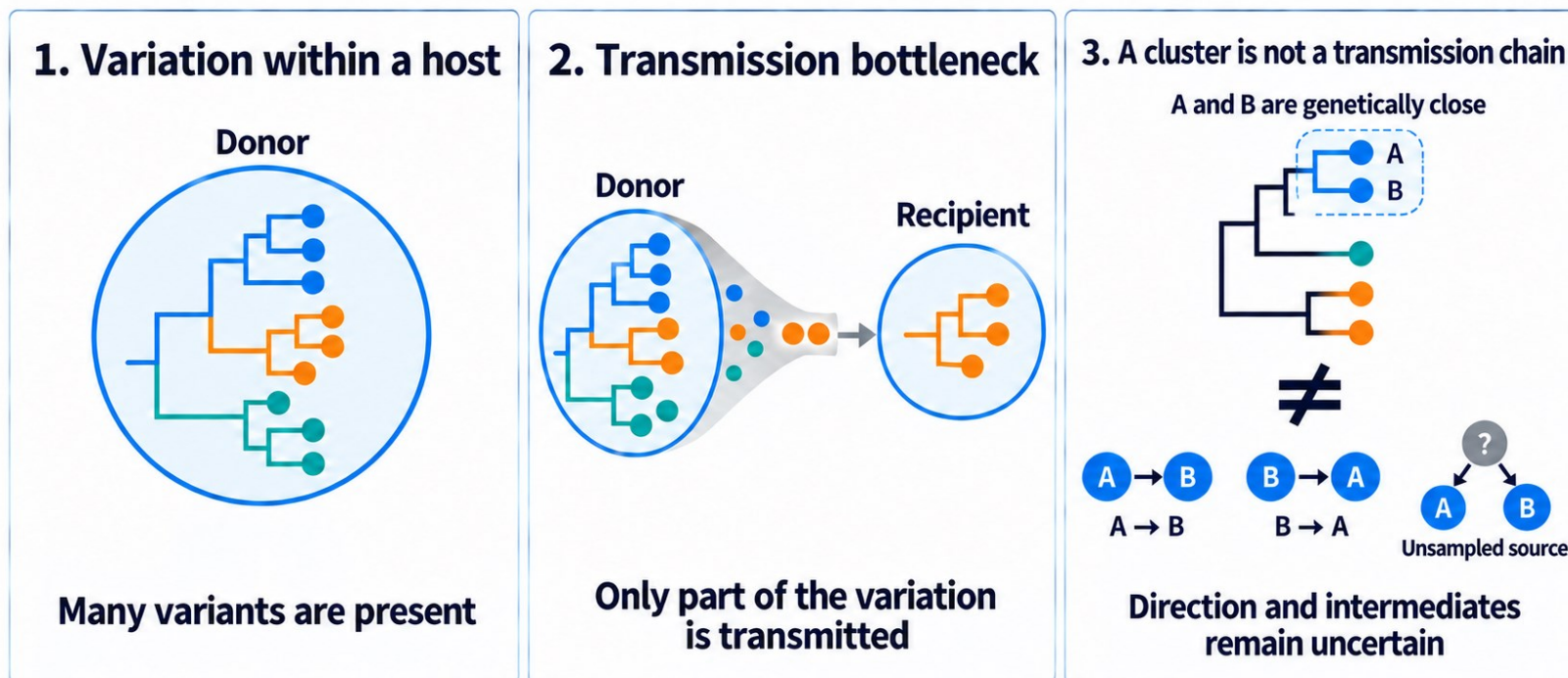


Viruses accumulate mutations during the epidemic process



Molecular Epidemiological Inferences

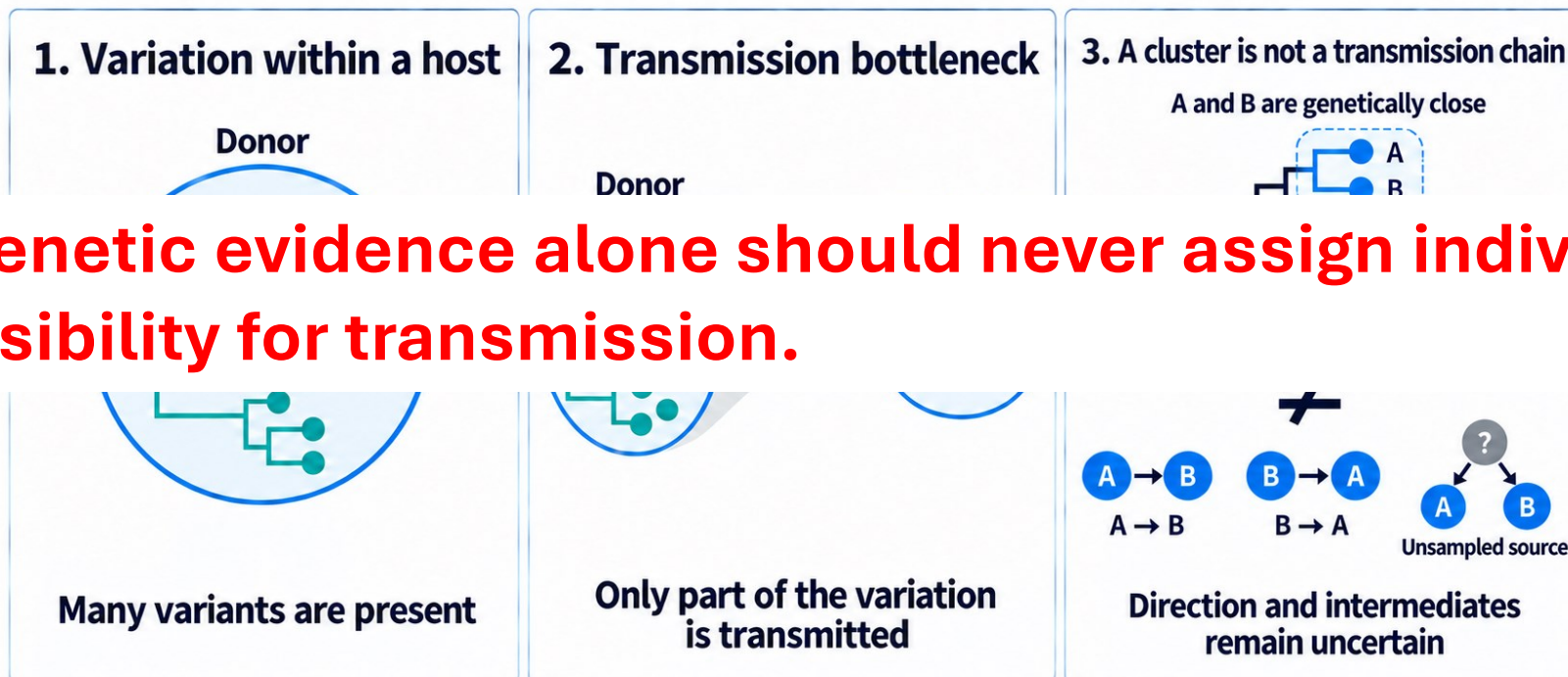
- Transmission: bottlenecks and contact structure move only part of that variation.



Genetic proximity can support epidemiological linkage—but cannot identify who infected whom.
Interpret with sampling dates, clinical history, contact data, and awareness of unsampled infections.

Molecular Epidemiological Inferences

- Transmission: bottlenecks and contact structure move only part of that variation.



Phylogenetic evidence alone should never assign individual responsibility for transmission.

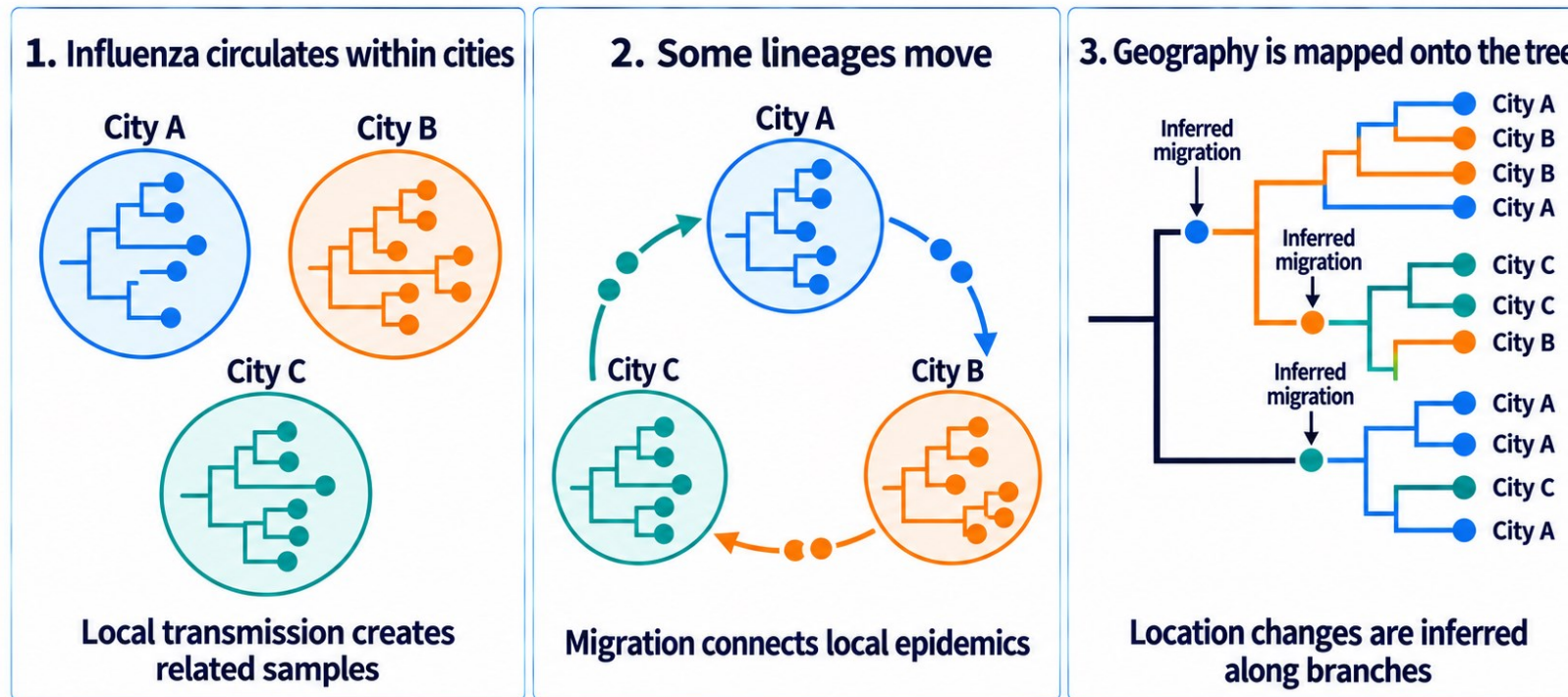
Genetic proximity can support epidemiological linkage—but cannot identify who infected whom.
Interpret with sampling dates, clinical history, contact data, and awareness of unsampled infections.

Structure redistributes diversity

- Migration introduces variants and can rescue lineages from local extinction.
- Source-sink dynamics and seasonal connectivity produce asymmetric lineage movement.
- A phylogeographic transition is an inferred ancestry change, not a directly observed transmission.

Structure redistributes diversity

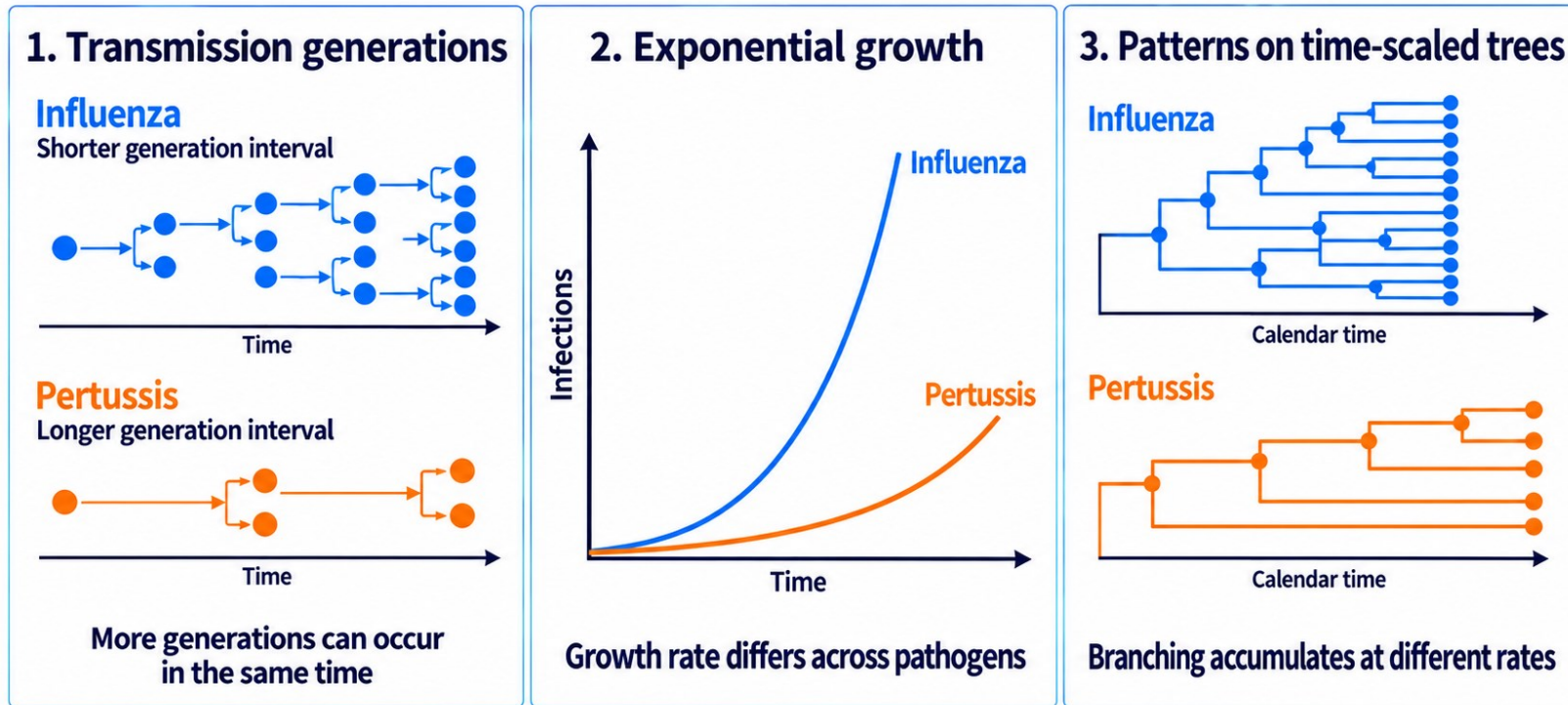
- Geographic, host, and ecological partitions reduce gene flow and permit divergence.



Colored branches represent inferred ancestral locations—not observed travel.
Sampling intensity and timing can alter inferred routes and apparent source–sink patterns.

Epidemic dynamics can leave signals in a phylogeny

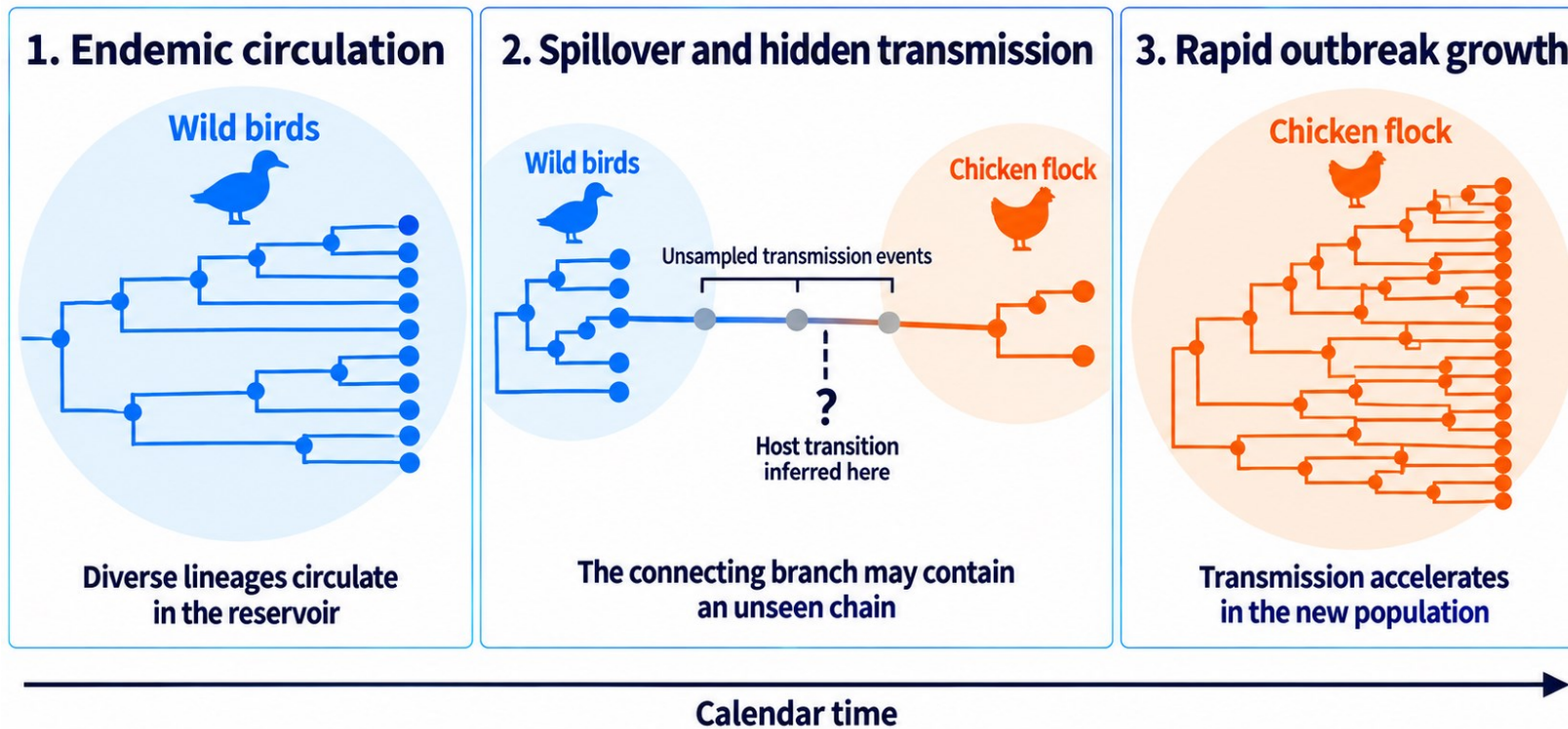
- Coalescent and birth–death models can estimate epidemic growth under explicit assumptions about sampling, population structure, and the molecular clock.



Epidemic growth rate depends on both reproductive success and the generation-interval distribution.
Tree shape also reflects sampling, population structure, and changes in transmission through time.

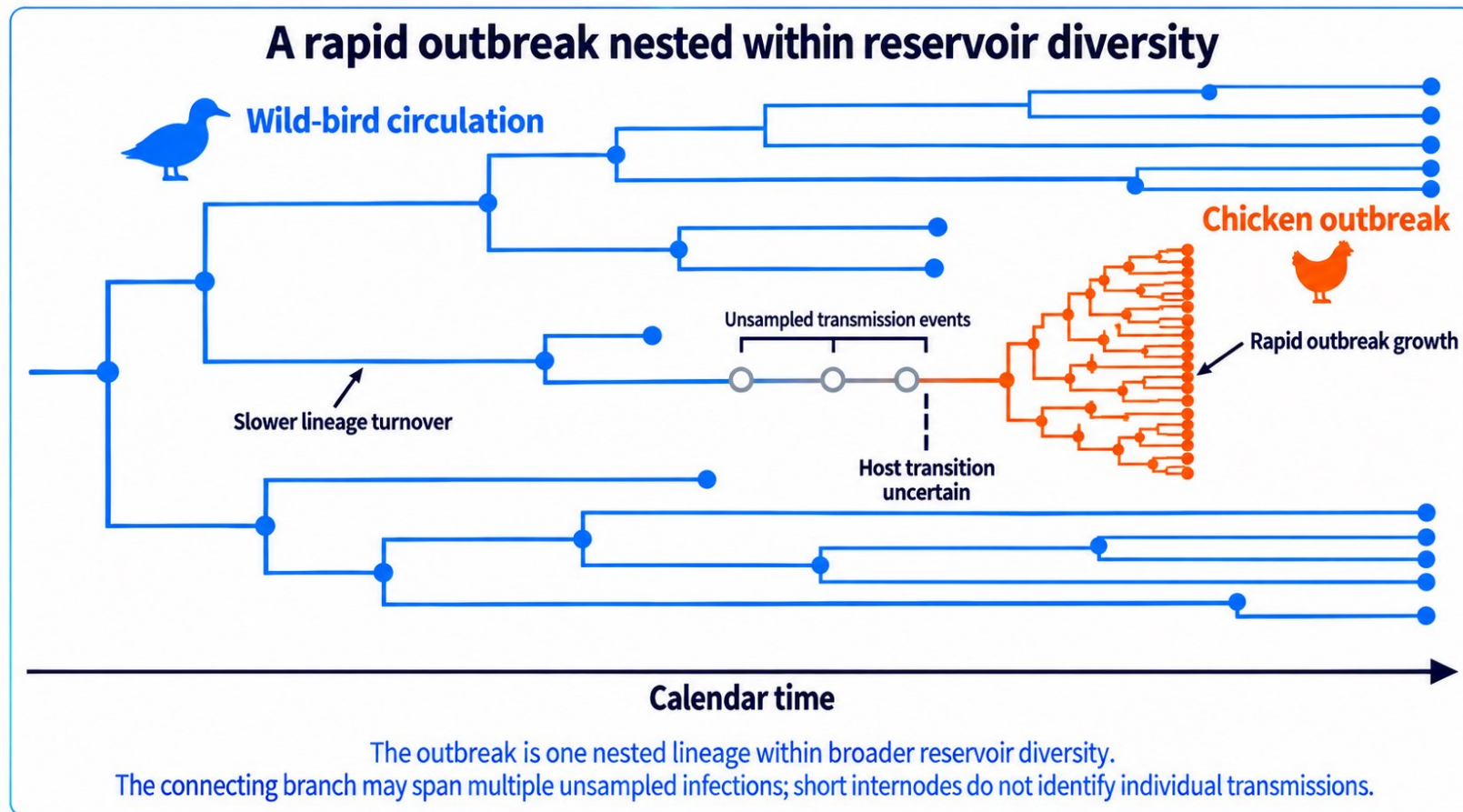
Epidemic growth depends on context

- Epidemic growth can change when a pathogen enters a population with different susceptibility, contact patterns, or generation intervals.

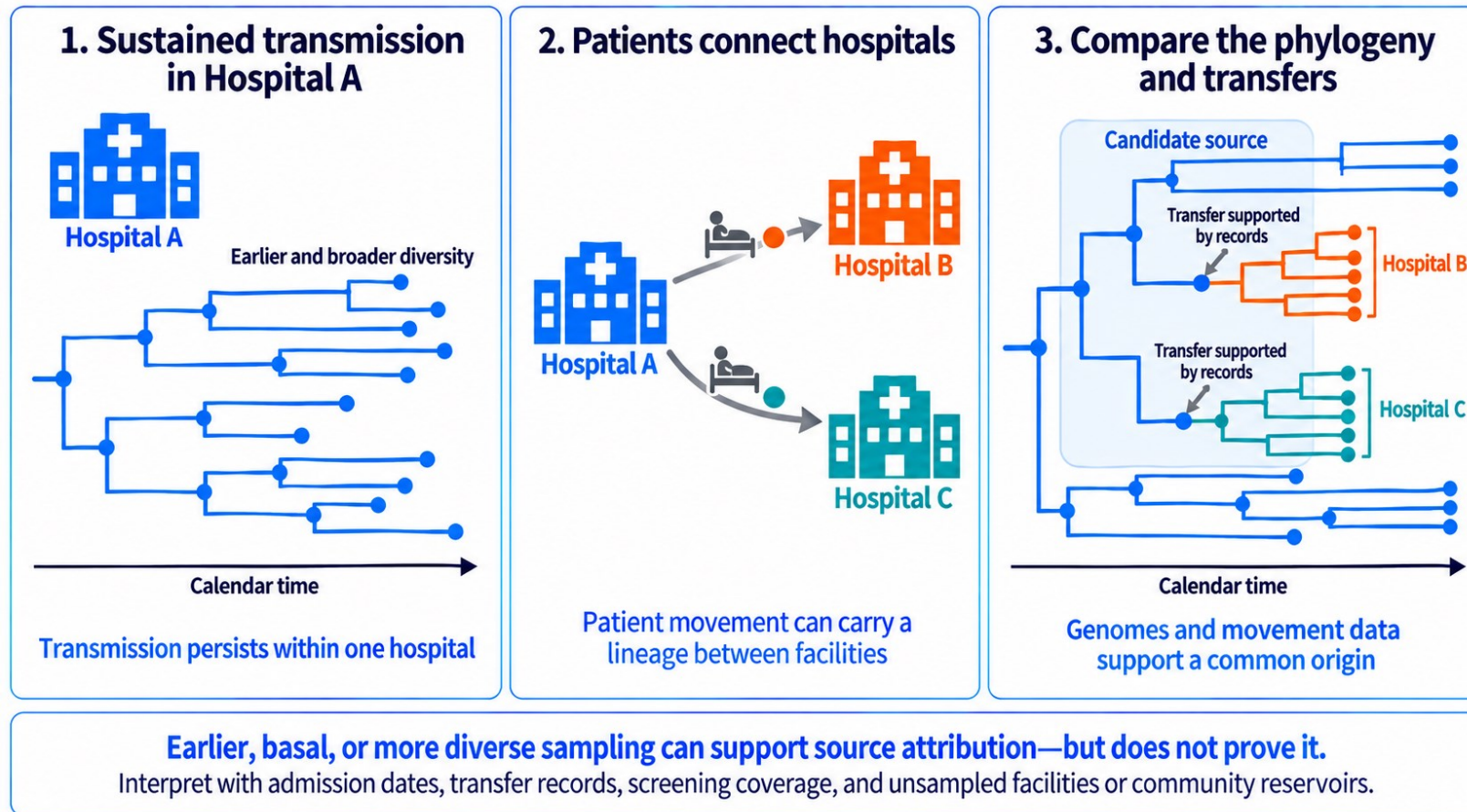


The branch leading to the outbreak is ancestry—not a single observed transmission.
It may include multiple unsampled infections, evolutionary changes, and delayed detection.

Spillover and transmission variation on a tree



Tracking a slow-moving pathogen

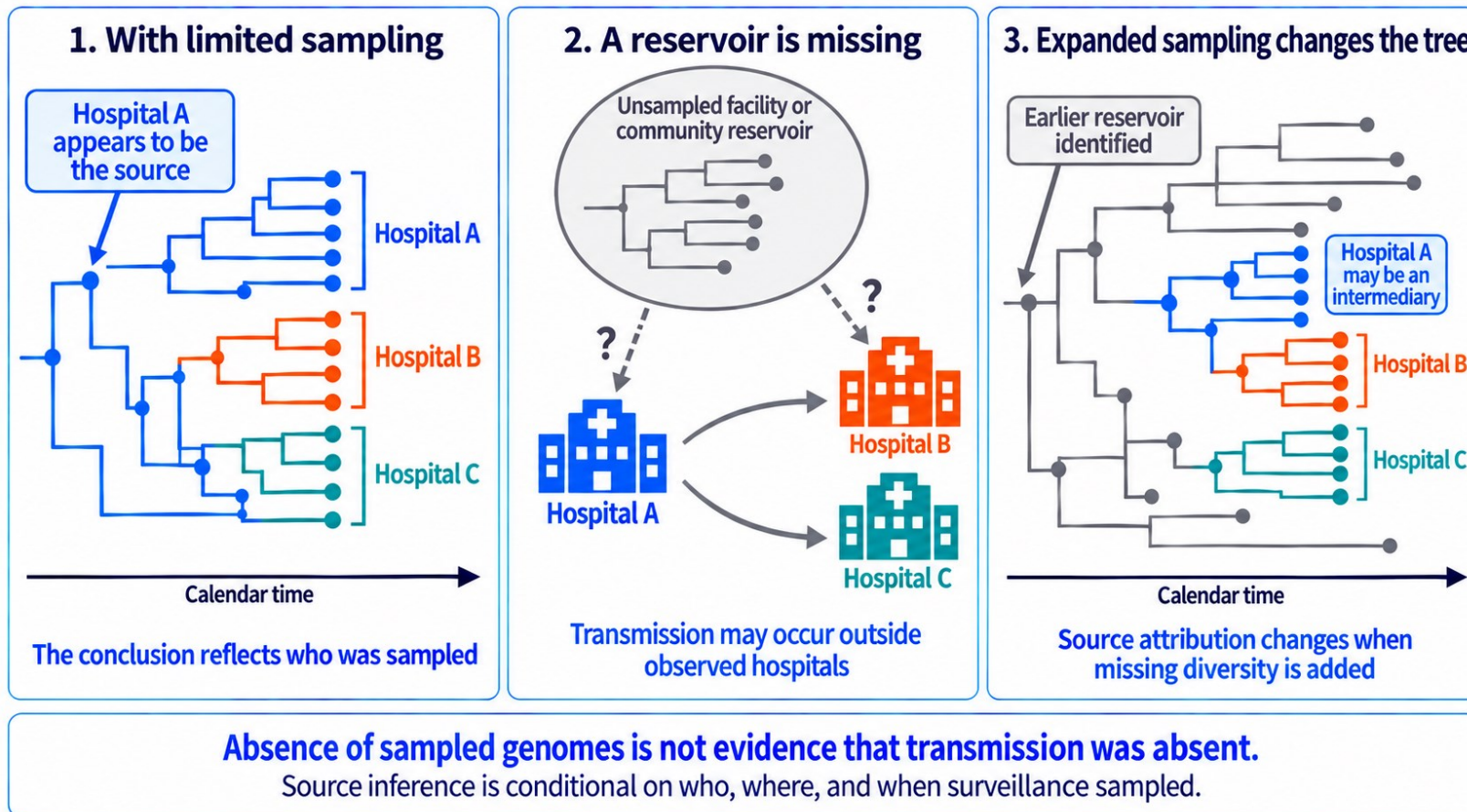


A strain phylogeny traces the bacterial lineage. It may not trace the history of a resistance gene or plasmid.

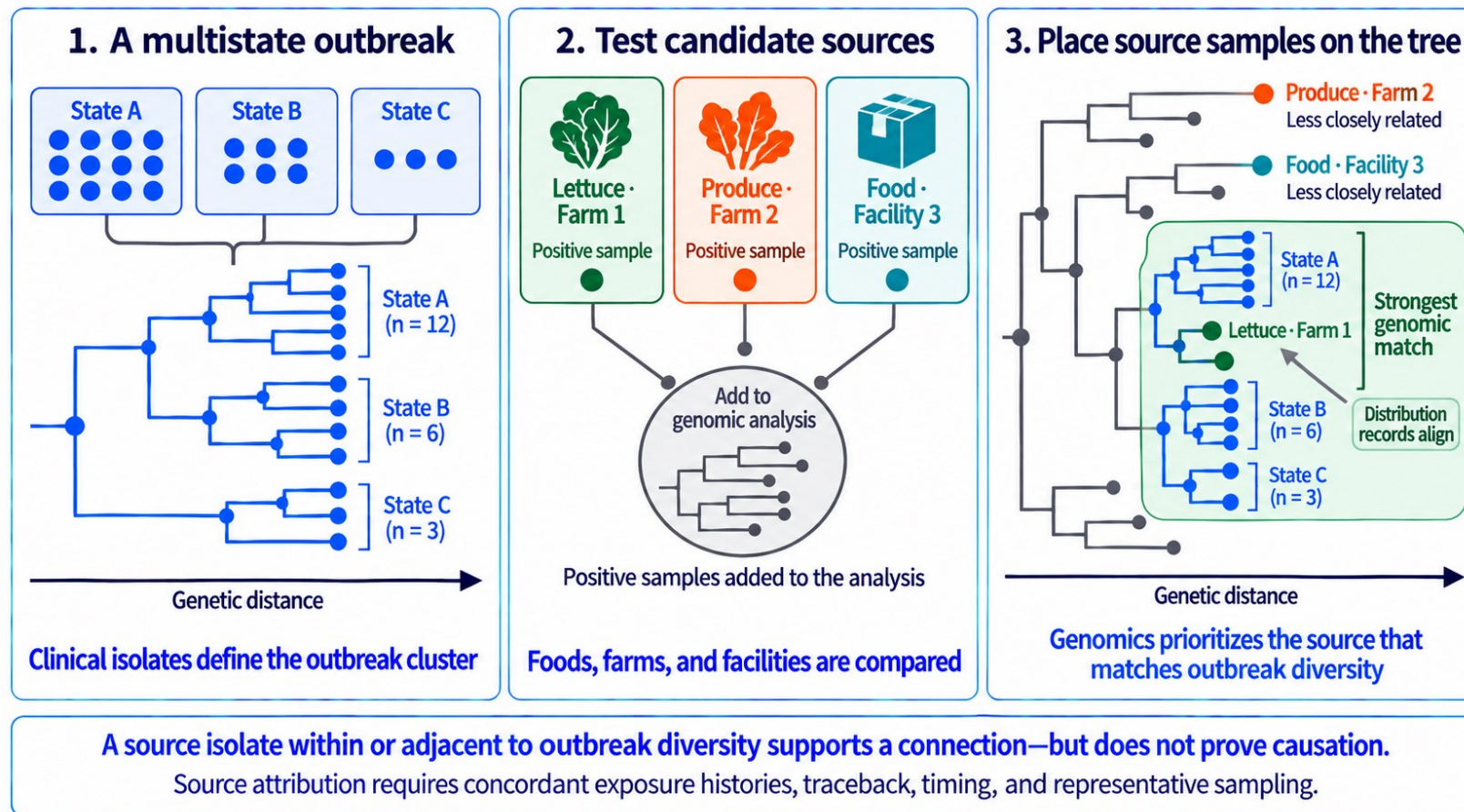
Sampling determines observed diversity

- Clinical ascertainment determines which infections enter surveillance.
- Specimen quality, assay thresholds, and sequencing protocols filter genomes.
- Uneven sampling through time, space, and host populations distorts frequencies and trees.
- Metadata completeness limits which evolutionary hypotheses can be tested.

What if we do not sample the source?



Contextual sequences can help elucidate potential outbreak sources



Fitness depends on scale

- A variant can replicate efficiently within hosts but transmit poorly between hosts.
- Immune escape can increase growth in immune populations while imposing costs elsewhere.
- Selection in one host species can oppose selection after a host jump.
- The relevant fitness measure must match the scale and outcome being explained.

A process-based interpretation

- What mechanism generated the variants?
- At what biological and epidemiological scale did frequencies change?
- Which demographic or sampling process could mimic selection?
- What additional phenotype, metadata, or sampling design would distinguish the explanations?

Key Takeaway

- A phylogeny is an inferred genealogy of sampled genomes.
- Similar patterns can arise from different evolutionary, epidemiological, and sampling processes.
- Credible inference combines explicit models with independent epidemiologic data.
- The tree constrains the possible explanations; it does not uniquely determine them.